

Supplemental material for: Source Grounding Verifies Retrieval and Not Interpretation in Automated Extraction from Construction Tender Documents

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This file is read on its own. It carries the schema the evaluation scores, the test cases of the numeric checks, the applicability of the arithmetic checks, the reconciliation of the two counts of the multi-country study, the campaigns that precede the canonical configuration, the configuration, reliability, metric, evidence and controlled-arm tables the evaluation rests on, and the tables, definitions and worked detail that the manuscript refers to but does not print.

Contents: Table S1, test cases for the numeric repair and the well-formedness test; Note S2, the locatability test and the input-selection rule; Table S3, observed error classes; Table S4, validation against published registry records; Table S5, selected fields with the engine that supplied each; Table S6, the twenty-two evaluated schema fields; Table S7, applicability of the arithmetic checks; Note S8, the two counts of the multi-country study and the severity of the 42; Note S9, campaigns preceding the canonical configuration; Table S10, configuration and scorer of every campaign; Table S11, operational reliability by arm; Table S12, metrics by configuration; Table S13, annotated evidence reaching the model input; Table S14, controlled arms; Table S15, limitations in current work and the mechanisms proposed; Table S16, summary-card fields; Note S17, the scoring criteria and the definition of every reported quantity; Table S18, populated, correct and unlocatable fields; Table S19, effect of removing each component; Table S20, label-free replication; Note S21, cross-provider evaluation of the accepted flat schema; Table S22, per-document cost calculation; Note S23, the three sensitivity analyses in full; Note S24, the merge rule, the error classes and the registry comparison.

Conventions. Every number here comes from the campaign named in Table S10. References to Table 1 to Table 4, to Fig. 1 to Fig. 6 and to named sections are to the manuscript. The canonical campaign is C-R2-4DOC, executed on 15 and 16 September 2026 on gemini-2.5-flash with temperature 0, top-p 1, a maximum output length of 8,192 tokens and the reasoning budget set to zero, and scored with the boundary-aware comparator.

Items S1 to S5 are the items the manuscript names directly, in the order in which it first names them. Items S6 to S24 follow in the order in which they appear here; they carry the reference listings, derivations and worked detail that the manuscript refers to but does not print, because the length limit of the journal does not allow them in the

body. No number in them is absent from the deposit and none contradicts the body. Table and figure numbering, captions and notes follow the conventions of the manuscript.

Table S1 gives the positive and negative cases for the text-layer repair and for the well-formedness test, with the class of each outcome.

Table S1. Test cases for the numeric repair and the well-formedness test.

Input	Output of the repair	Class	What the case shows
importe de 40 .000,00 EUR	importe de 40.000,00 EUR	correct repair	the observed text-layer defect
importe de 35.000 ,00 EUR	importe de 35.000,00 EUR	correct repair	a space before the decimal comma
importe de 8 0.000,00 EUR	importe de 80.000,00 EUR	correct repair	a digit split from a numeric group
importe de 3 5 .000,00 EUR	importe de 35.000,00 EUR	correct repair	the fixed-point loop is needed
total 1.161.820 ,80 EUR	total 1.161.820,80 EUR	correct repair	a large amount with a split decimal
Anexo 3 000 personas	unchanged	correctly left alone	a plain three-digit group after a digit
lotes 2 y 3 del contrato	unchanged	correctly left alone	adjacent single digits
plazo 24 meses y 3 dias	unchanged	correctly left alone	durations carrying no separator
base 100.000,00 IVA 21.000,00	unchanged	correctly left alone	two adjacent complete amounts
clausula 12 . El adjudicatario	clausula 12. El adjudicatario	harmful repair	a sentence-final full stop absorbed
apartado 5 , letra b)	apartado 5, letra b)	harmful repair	a comma after an item number absorbed
item 7 1.234,00 y item 8	item 71.234,00 y item 8	harmful repair	two adjacent numbers merged into a fabricated amount

Input	Output of the repair	Class	What the case shows
000,00 EUR, 00 EUR, 48 EUR, 77 EUR, 15 EUR, 25 EUR	withheld by the well-formedness test	correct rejection	every truncation observed in the two corpora
0,00 EUR and 0 EUR	withheld	wrongly withheld	a zero amount where a zero is permitted
950 EUR	withheld	wrongly withheld	a legitimate whole-unit amount below one thousand
1'234.56 EUR	withheld	wrongly withheld	an amount grouped with an apostrophe

Analysis of Table S1 shows that the repair behaves as described on the defect it was written for and on the constructions it is meant to avoid, and that it has three failure modes the description does not predict, of which the third fabricates an amount that the locatability test then admits. The well-formedness test rejects every truncation observed in either corpus, and withholds three classes of legitimate value, so the assumption that a genuine tender figure always carries a separator is a jurisdictional convention and not a rule of the domain.

Note S2 states as pseudocode the exact acceptance condition of the source-grounding gate and the rule that selects the passage sent to the model. The condition stated in the manuscript, that a reference and an excerpt be located together within a bounded window, describes the deployed prototype; the evaluation harness applies the two functions below, which test the value against the whole normalised document and use no window. A value carrying several numbers is admitted when any one of them occurs in the source.

Note S2. The locatability test and the input-selection rule.

```
normalise(s):
```

```
    s <- NFKD(s); drop combining marks; collapse whitespace; trim
    return lowercase(s)
```

```
grounded_strict(value, source):          # the canonical gate
```

```
    if value is empty:    return false
```

```
    N <- numeric tokens of value: a grouped amount such as 1.234,56,
```

```
    or a count followed by MESES, MES, SEMANAS, ANOS or DIAS
```

```

52     if N is not empty:      return some n in N with normalise(n) in source
53     v <- normalise(value)
54     if length(v) < 4:      return v occurs in source
55     return the first min(18, length(v)) characters of v occur in source
56
57 grounded_robust(value, source):      # post-hoc variant
58     as grounded_strict up to v
59     T <- the tokens of v of length 4 or more
60     if T is empty:      return the first min(18, length(v)) characters of v
61 occur in source
62     return every t in T occurs in source
63
64 select_input(text):
65     i <- offset of the LAST line beginning "ANEXO I"
66     if there is none:      return text
67     j <- offset of the first heading "ANEXO II" .. "ANEXO X" after i + 10
68     if there is none: j <- min(length(text), i + 120000)
69     focus <- text[0 : min(12000, i)] + "[...]" + text[i : j]
70     if length(focus) >= 8000 and length(focus) < 0.8 x length(text):
71         return focus
72     return text      # otherwise the whole document
73
74     the caller then truncates the result to 120,000 characters
75
76     Three properties of the input-selection rule are consequences rather than choices. The last line beginning
77 ANEXO I is taken, not the first, which is how an entry in a table of contents is skipped and also how a cross-
78 reference late in the document moves the window. Competing annexes are not handled: whichever of ANEXO II to
ANEXO X occurs first ends the window, whatever it contains. A cross-reference from the body to an annex outside

```

the window is not followed, and no test is made that the selected window contains the fields being sought. Table S13 reports the share of annotated evidence that reaches the model under this rule and under two alternatives.

Table S3 catalogues the residual errors of the canonical configuration with the mechanism and the check that detects each.

Table S3. Observed error classes and the check that detects each.

Error class	Mechanism	Example	Detected by
Numeric truncation	A space inside a number in the text layer leaves a well-formed fragment that a pattern matches	“40 .000,00 €” surfaced as “000,00 €”	Not the grounding gate, since the fragment is verbatim; only the text-layer repair and the well-formedness check
Paraphrase rejection	The model returns a phrase where the source carries the head word, and the strict prefix test rejects a correct value	Contract type returned as a phrase	No check; it is a false rejection, recovered by the relaxed locatability test of Note S23
Field misassignment	A value present verbatim in the document is attached to the wrong field	Contracting authority; definitive guarantee	Neither gate; detectable only against labels, and reported as the false-positive column of Table 3 of the manuscript
Precedence override	A correct language-model value is displaced by a deterministic value during the merge	Contract type on two documents	No check; field-level routing is identified as future work

Analysis of Table S3 shows that no error class is stochastic and that two of the four are invisible to both verification gates. One recurring error concerns a budget figure in the prose specification from the regional health service, where the relevant amount appears in a sentence alongside a closely related figure of the same type and the value surfaced is the neighbouring one; this is an instance of the limit of source grounding stated in the manuscript. Beyond it, most of the raw-coverage shortfall is concentrated in the model and framework documents, where many target fields are deferred to later annexes and are absent by design.

Table S4 compares the extracted values with the structured record the contracting platform publishes alongside each specification, an independent reference created by the contracting authority and never inspected during development.

Table S4. Validation against published registry records.

Field	Tenders with a published reference	Extra cted	Agreement and completeness	Status
Procedure type	26	24	24 of 24 emitted (100 %); 24 of 26 available (92.3 %)	scored

Field	Tenders with a published reference	Extracted	Agreement and completeness	Status
Base budget	26	15	not applicable	not scored

Analysis of Table S4 shows complete agreement among the values that were emitted together with recovery of 24 of the 26 available references, so a conditional agreement of 100 percent coexists with a completeness of 92.3 percent; both are reported, because the first alone would overstate what the comparison establishes. The budget field is not scored, because the platform publishes three distinct amounts and the harvested manifest retained one of them without recording which. On the ten tenders where both a published amount and a tax-inclusive extraction exist, six ratios are 1.00 and two are 1.21, which is what a manifest mixing tax-inclusive with tax-exclusive amounts would produce, and the remaining two match neither; repairing the reference requires re-harvesting the platform records. The reference is a silver standard rather than an absolute one, since the published record is itself keyed in by hand, so a disagreement locates a discrepancy without establishing on which side it lies, and the comparison does not replace independent re-annotation of the twenty verified values, because the platform publishes none of the solvency, classification or guarantee thresholds that motivate this work. Table S20 and Table S4 both come from campaign R-POST of August 2026 on the twenty-six-document corpus and neither carries a scorer, that corpus having no reference values.

Table S5 is a worked example of the output of the workflow on real documents. The first six rows are taken from a municipal works specification provided as a public supplementary document, the construction of a multi-functional civic building, on which the deterministic engine populates every field shown. The last two rows are taken from Pliego B, the prose specification of the evaluation corpus, and are fields the deterministic engine leaves empty and the language model supplies; they are included because a table drawn only from the standard template would show one engine of the two.

Table S5. Selected fields, with the engine that supplied each.

Field	Extracted value	Reference	Method
Contract type	Works (obras)	Annex I, item 1	rule
Procedur e	Open (abierto)	Annex I, item 1	rule
Estimated value	5,532,479.98 €	Annex I, item 10	rule
Base budget	5,532,479.98 €	Annex I, item 10	rule

Field	Extracted value	Reference	Method
(excl. VAT)			
VAT	21 % (1,161,820.80 €)	Annex I, item 10	rule
Total budget (incl. VAT)	6,694,300.78 €	Annex I, item 10	rule
Technical solvency (Pliego B)	Relación de obras ejecutadas en los últimos cinco años, referidas al Grupo: I Instalaciones eléctrica; Subgrupos: 1 Alumbrados, iluminaciones y balizamientos luminosos.	Criterios de solvencia técnica o profesional	language model, admitted by the gate
Mandator y insurance (Pliego B)	Seguro de responsabilidad civil con el límite mínimo de indemnización de la póliza de 10.000 €.	Clause k, obligations of the successful tenderer	language model, admitted by the gate

Analysis of Table S5 shows the division of labour that the ablation measures. Every value carries a reference to the clause from which it was read, and the budget figures reconcile under the validation gate. For a specification that follows the standard template the deterministic engine populates the monetary record on its own, while on the prose specification the two solvency and insurance fields reach the record only through the language model and the gate. Neither part surfaces a value that fails the locatability test.

Table S6 prints the twenty-two fields the evaluation scores, with the type of each and the states it may take when no value is present.

Table S6. The twenty-two evaluated schema fields.

Field	Type	Meaning	Missing states and cardinality
organo_contratacion	text	contracting body	empty when not stated; one value, no lot scope
numero_expediente	text	file reference	empty when not stated; one value
objeto_contrato	text	object of the contract	empty when not stated; one value
presupuesto_base_sin_iva	amount	base budget excluding value added tax	empty when deferred to an annex; one value, no lot scope
presupuesto_base_con_iva	amount	base budget including value added tax	empty when deferred; one value, no lot scope

Field	Type	Meaning	Missing states and cardinality
valor_estimado_contrato	amount	estimated contract value	empty when deferred; one value
plazo_ejecucion	duration	execution period	empty when deferred; one value, no per-lot period
plazo_presentacion_ofertas	date or duration	deadline for offers	empty when fixed by the notice; one value
lugar_presentacion	text	place or platform of submission	empty when not stated; one value
garantia_provisional	amount or percentage	provisional guarantee	empty when none is required; one value, no scope
garantia_definitiva	amount or percentage	definitive guarantee	empty when deferred; one value, no scope
solvencia_economica	text	economic solvency requirement	empty when deferred; one value, alternatives flattened
solvencia_tecnica	text	technical solvency requirement	empty when deferred; one value, alternatives flattened
clasificacion_empresarial	text	business classification	empty when not required; one value
procedimiento	category	award procedure	empty when not stated; one value
tramitacion	category	processing type	empty when not stated; one value
tipo_contrato	category	contract type	empty when not stated; one value
iva	percentage	value added tax rate	empty when deferred; one value, no per-item rate
ofertas_anormalmente_bajas	text	abnormally low offer rule	empty when not stated; one value
plazo_garantia	duration	guarantee period	empty when not stated; one value
seguros_obligatorios	text	mandatory insurance	empty when not required; one value

Field	Type	Meaning	Missing states and cardinality
presentacion_electronica	boolean or text	electronic submission required	empty when not stated; one value

Analysis of Table S6 shows that every field is single-valued and that no field carries a scope. A tender divided into lots, an alternative qualification condition and a guarantee stated for part of the contract are therefore representable only by discarding information, and the annotation protocol described in the manuscript records such cases as ambiguous rather than forcing them into a single value. Three further members of the implementation schema, the award criteria, the subcontracting permission and the price-revision flag, are not scored and are not listed here.

Table S7 reports how often each arithmetic check was applicable and how many mismatches it found, on both evaluated corpora.

Table S7. Applicability of the arithmetic checks on both corpora.

Check	Applicable, four documents	Mismatches, four documents	Applicable, twenty-six documents	Mismatches, twenty-six documents
base plus value added tax equals the total	0 of 4	0	0 of 26	0
tax equals the base times the rate	0 of 4	0	0 of 26	0
estimated value is at least the base budget	0 of 4	0	0 of 26	0
monetary value is a well-formed amount	2 of 4	0	23 of 26	5

Analysis of Table S7 shows that the three monetary identities were never once applicable. All three take the base budget excluding value added tax as an operand, and that field was surfaced zero times in either corpus, which is why their applicability is zero throughout. The tolerance is relative and set at 2 percent. Lot-level and contract-level amounts are not distinguished, because the schema of Table S6 carries no lot identifier. All five mismatches of the fourth row were produced by the deterministic engine and none by the model. The component is therefore reported as built and not exercised, and the accuracy of the workflow is not attributable to it.

Note S8 reconciles the two counts the manuscript reports for the same failure, and gives the distribution of the error factor over all 42 silently wrong values.

Note S8. The two counts of the multi-country study, and the severity of the 42.

The manuscript reports two numbers for one failure and they count different things. Classified by the number the shipped parser returns, the 87 published amounts divide into 0 correct, 42 silently wrong and 45 refused, a refusal being any non-positive result. Classified by the substring the parser matched, which is what the grounding predicate is actually offered, 86 fragments were surfaced and 86 were admitted. The arithmetic closes in both directions: the 86 are the 42 that yield a usable but wrong number plus the 44 whose fragment reads zero, and the 45 refusals are those same 44 plus the single amount printed without grouping, on which the parser matches nothing at all. Of the 87 amounts, 86 are printed space-grouped and one plain, and the plain one is the amount that matches nothing.

The consequence for the paper is carried by the second count. A refusal is not the benign outcome the word suggests, because 44 of the 45 surfaced a fragment reading zero into a monetary field and the gate admitted every one of them. The check that stops those 44 is the well-formedness test of the methodology, which withholds a bare fragment carrying leading zero digits, and it is not a test of locatability.

The severity of the 42 was measured over all of them and not over the ten worked examples the harness prints. The error factor, the published reference divided by the number the shipped parser returns, has a minimum of 39.67 on an Austrian notice printing 14 362,00 that the parser reads as 362, a median of 6,179.89, and a maximum of 1,585,419.76 on a Danish notice printing 567 897 358,20 that it reads as 358.20. None of the 42 falls below a factor of ten, six lie between ten and a thousand, and thirty-six are a thousand or more. Earlier versions of this work quoted the range as 135 to 11,893; those are the smallest and largest of the ten printed worked examples, not of the 42 measured values, and the figure is corrected throughout this version.

Note S9 records the campaigns executed before the sampling parameters of the canonical configuration were fixed, and the replication campaign executed before the text-layer repair. No figure in the manuscript comes from any of them, except the comparison drawn in Figure 4.

Note S9. Campaigns preceding the canonical configuration.

Three campaigns on the four labelled documents were executed before the sampling parameters were fixed, one pinned campaign was executed in August 2026 on the same four documents, and one replication campaign was executed before the text-layer repair. Their identifiers, configurations and archives are listed in Table S10.

The August 2026 pinned campaign, C-PIN-4DOC-B, ran the configuration of the canonical campaign on the same four documents and was scored with substring containment, the comparator the harness shipped with. Under that comparator it returns 16.0 of 20 for the full workflow and a precision of 88.9 %, against 17.0 of 20 and 94.4 % under the boundary-aware comparator. The whole of the difference is one labelled slot, the definitive guarantee of Pliego A, where the reference value is 5 % and the surfaced value is 5 % del precio final ofertado: containment cannot cross the space between the digit and the percent sign, and scores as wrong a value that is right. The September campaign reproduces the August one exactly when it is scored the August way, so the two campaigns agree and only the comparator differs.

The three unpinned four-document campaigns left the sampling temperature, the maximum output length and the reasoning budget at the provider defaults. For Gemini 2.5 Flash the default sampling temperature is 1.0, and the provider publishes no exact default for the other two, which were therefore not fixed. Their campaign means for the full workflow were 17.0, 16.2 and 16.4 of 20; pooled over their thirty runs the grand mean is 16.5 of 20, with a within-campaign standard deviation of 1.12 and a between-campaign standard deviation of 0.42. The dispersion is attributable to the default temperature, and it quantifies the reproducibility risk of querying a hosted model without pinning its configuration. The paired gate cost of 1.30 plus or minus 0.27 correct values is a property of two of these campaigns, and the ranges of 15 to 19 correct values and 44 to 64 populated slots belong to them as well.

The replication campaign executed before the text-layer repair, R-PRE, gave 402.0 plus or minus 13.1 slots for the language model alone with 34.0 plus or minus 2.5 percent of its values failing the locatability test, 417.7 plus or minus 14.1 for the ungated hybrid at 29.7 plus or minus 2.9 percent, and 294.7 plus or minus 5.2 for the full workflow. It is the campaign against which Figure 4 of the manuscript compares.

Table S10 lists every campaign from which any number in this paper is taken, with the scorer each was read under, so that each table and figure can be traced to one configuration and one comparator.

186 Table S10. Configuration and scorer of every campaign.

Campaign	Archive	Model	Sampling	Reasoning	Runs	Pre-processing	Scorer	Reported in
C-R2-4DOC (canonical)	data/runs/2026-09/results.json	gemini-2.5-flash	temperature 0, top-p 1, max output 8,192	budget 0	10 per arm	after the repair, except the no-repair arm	boundary-aware	Table 2, Table 3, Table S18, Table S19, Table S3, Table S5, Table S22, Table S11, Table S12, Table S13, Table S14, Fig. 2, Fig. 3
C-PIN-4DOC-B	results/campaign-logs/4doc-10runs.log	gemini-2.5-flash	temperature 0, top-p 1, max output 8,192	budget 0	10	after the repair	containment	Note S9 only
C-PIN-4DOC-A	results/campaign-logs/4doc-pinned-a.log	gemini-2.5-flash	temperature 0, top-p 1, max output 8,192	budget 0	10	before the repair	containment	nowhere; retained for comparison
C1, C2, C3	results/campaign-logs/4doc-unpinned-c1.log, c2 and c3	gemini-2.5-flash	provider defaults (temperature 1.0)	not recorded	10 each	before the repair	containment	Note S9 only

Campaign	Archive	Model	Sampling	Reasoning	Runs	Pre-processing	Scorer	Reported in
C-DS-4DOC, C-SCHEMA	results/campaign-logs/crossprovider.log	deepseek-chat (moving alias, no snapshot identifier)	temperature 0, max tokens 4,000, JSON object	not applicable	5 and 1	after the repair	containment	Note S21, labelled as a second provider
R-POST	results/pooled-26doc-replication.txt, over the three transcripts in results/campaign-logs/	gemini-2.5-flash	temperature 0, top-p 1, max output 8,192	budget 0	3	after the repair	no scorer; label-free	Table S20, Table S4
R-PRE	results/campaign-logs/26doc-pre-repair.log	gemini-2.5-flash	not recorded	not recorded	3	before the repair	no scorer; label-free	Note S9 and Fig. 4
TED-87	analysis/numeric-failure/, with its outputs in results/numeric-failure/	no model call	not applicable	not applicable	deterministic	not applicable	exact match to the published value	Table 4, Fig. 5, Fig. 6

Analysis of Table S10 shows that the canonical campaign and the August pinned campaign differ in no parameter of the configuration and only in the comparator, that the three unpinned campaigns differ in the sampling temperature, and that the two replication campaigns differ in the pre-processing version. Four fields that a reader would want are

absent from every campaign before C-R2-4DOC and are recorded here as absent: the execution date, which survives only as a file modification time; a gate version identifier, the gate being distinguished only by which function the harness calls; a pre-processing version identifier, the repair being an uncommitted change to the cleaning library; and the token usage of any campaign, which the archived harness never read. C-R2-4DOC records all four.

Table S11 reports the request-level outcome of every arm of the canonical campaign, which the archived campaigns did not record at all.

Table S11. Operational reliability by arm, campaign C-R2-4DOC.

Arm	Factor changed against the control	Document calls	Valid and parseable	Usable records	Retries per call	Latency, median and p90	Tokens, prompt and output
control	none; the canonical configuration	40 over 10 runs	100 % valid, 100 % parseable	100 %	0.10, none exhausted	20.0 s and 24.1 s	23,029 and 3,914
schema_flat	a flat schema the endpoint accepts, in place of schema-less decoding with the normalizer	20 over 5 runs	100 % valid, 100 % parseable	100 %	0.00, none exhausted	5.4 s and 9.0 s	23,029 and 1,103
input_fulltext	the whole cleaned text in place of head and annex	40 over 10 runs	100 % valid, 100 % parseable	100 %	0.78, none exhausted	31.6 s and 43.2 s	88,502 and 5,155
input_passage	cue-driven per-field passages in place of head and annex	40 over 10 runs	100 % valid, 100 % parseable	100 %	1.57, none exhausted	22.5 s and 35.9 s	25,495 and 4,372

Arm	Factor changed against the control	Document calls	Valid and parseable	Usable records	Retries per call	Latency, median and p90	Tokens, prompt and output
no_repair	the joinSplitNumbers repair disabled	40 over 10 runs	100 % valid, 100 % parseable	100 %	1.45, none exhausted	17.8 s and 32.6 s	23,066 and 3,545
docie_baseline	generic key-value extraction with no schema, normalizer, gate or rules	20 over 5 runs	100 % valid, 75 % parseable	65 %	0.25, none exhausted	16.7 s and 44.7 s	21,602 and 4,207

Analysis of Table S11 shows that the extraction quality reported everywhere else in this paper rests on requests that succeeded: across the six arms every request was valid, and every response was parseable except on the baseline arm. The two quantities that vary are latency, which is three times the canonical figure when the whole document is sent and a third of it when the flat schema removes the normalizer from the path, and truncation, which occurs on one arm only. Every arm ran the canonical sampling configuration and differs from the control in one factor, and every arm is complete at ten runs, or five for the two whose outcome is accuracy alone. The docie_baseline arm is the only one at which a response was truncated at the output limit, on 5 of its 20 document calls, which is why its usable-record completion is below 100 percent.

Table S12 reports the metrics per configuration, from the canonical campaign, with the containment figure beside them.

Table S12. Metrics by configuration, campaign C-R2-4DOC.

Configuration	Precision	Recall	F1	Abstention	Incorrect emission	Raw coverage of 88	Locatability pass rate	Correct of 20 under containment
Deterministic engine only	100.0 %	65.0 %	78.8 %	7.0	0.0	13.0	100.0 %	13.0

Configuration	Precision	Recall	F1	Abstention	Incorrect emission	Raw coverage of 88	Locatability pass rate	Correct of 20 under containment
Language model alone	89.2 % ± 0.3	82.5 % ± 2.6	85.7 % ± 1.6	1.5 ± 0.5	2.0	57.0 ± 2.8	81.6 %	15.5 ± 0.5
Language model under the gate, no rules	87.9 % ± 0.4	72.5 % ± 2.6	79.4 % ± 1.7	3.5 ± 0.5	2.0	46.5 ± 3.7	100.0 %	13.5 ± 0.5
Hybrid, ungated	95.0 %	95.0 %	95.0 %	0.0	1.0	58.5 ± 2.3	82.1 %	18.0
Full workflow (canonical)	94.4 %	85.0 %	89.5 %	2.0	1.0	48.0 ± 3.2	100.0 %	16.0
Relaxed gate (post-hoc, in-sample)	94.7 %	90.0 %	92.3 %	1.0	1.0	45.0 ± 2.1	93.3 %	18.0

Analysis of Table S12 shows precision, F1, abstention and incorrect emission for every configuration. The deterministic engine emits on thirteen of the twenty reference fields and is correct on all thirteen; the full workflow emits on eighteen and is correct on seventeen. The gap between recall and coverage is wide for every language-model configuration, which is the reason raw coverage is kept as a separate descriptive quantity rather than reported as performance. The first eight columns are scored with the boundary-aware comparator and the last with substring containment, which is the only column of this table that is not canonical and is printed so that the two can be compared. Applicable-field coverage is absent, because no per-document list of applicable fields exists. The locatability pass rate of the relaxed gate is measured against the strict test, not against its own.

Table S13 reports the share of annotated evidence that the input-selection rule places inside the model request, against two alternative rules.

218 Table S13. Annotated evidence reaching the model input.

Document	Pages	Source characters	Head and annex (current rule)	Whole cleaned text	Field-specific passages
Pliego A	152	504,128	8 of 8, 23.8 % of the text	8 of 8	7 of 8
Pliego B	26	78,326	7 of 7, 100 % of the text	7 of 7	7 of 7
Pliego C	122	387,912	2 of 3, 3.2 % of the text	3 of 3	3 of 3
Pliego D	105	414,857	2 of 2, 28.9 % of the text	2 of 2	2 of 2
All four	405	1,385,223	19 of 20, 330,584 characters sent	20 of 20, 1,385,223 sent	19 of 20, 375,099 sent

219 Analysis of Table S13 shows that the current rule withholds one of the twenty annotated values from the model and
220 that the document it starves is Pliego C, of which it sends 3.2 percent. Supplying the whole cleaned text places every
221 annotated value inside the request at four times the input size; passage selection places nineteen at a size comparable
222 to the current rule but loses a different one, on Pliego A, to the truncation cap. This measurement bounds what the
223 model could recover; Table S14 reports what each rule then recovers.

224 Table S14 reports the controlled arms, each changing exactly one factor against a control that reproduces the canonical
225 configuration.

226 Table S14. Controlled arms, campaign C-R2-4DOC.

Arm	Factor changed against the control	Raw coverage of 88	Correct of 20	Precision	Model values passing locatability
Control	none; the canonical configuration	48.0 ± 3.2	17.0	94.4 %	81.6 %

Arm	Factor changed against the control	Raw coverage of 88	Correct of 20	Precision	Model values passing locatability
Flat structured schema	a flat schema the endpoint accepts, in place of schema-less decoding with the normalizer	49.4 $\pm$ 0.9	18.0	94.7 %	90.5 %
Whole cleaned text	the whole cleaned text in place of head and annex	61.4 $\pm$ 0.5	18.0	94.7 %	88.1 %
Field-specific passages	cue-driven per-field passages in place of head and annex	58.6 $\pm$ 1.6	19.0	95.0 %	88.7 %
No text-layer repair	the joinSplitNumbers repair disabled	47.8 $\pm$ 1.5	17.0	94.4 %	82.7 %
Generic extraction baseline (the baseline alone)	generic key-value extraction with no schema, normalizer, gate or rules	19.0 $\pm$ 6.7	4.6 $\pm$ 0.5	82.0 % $\pm$ 1.8	100.0 %

Analysis of Table S14 shows that three of the five configurations recover more reference values than the canonical one. Passage selection reaches 19.0 of 20 at an input size comparable to the canonical rule, the flat structured-output schema 18.0 of 20 on the fewest output tokens of any arm, and the whole cleaned text 18.0 of 20 at nearly four times the prompt tokens. Disabling the text-layer repair changes nothing measurable, at 17.0 of 20 against 17.0. The generic baseline, which has none of this paper apparatus, reaches 4.6 $\pm$ 0.5 of 20. The canonical configuration is therefore not the best configuration measured here, and the manuscript states why it is nonetheless the one this paper reports. Every row but the last is the full workflow of one arm of campaign C-R2-4DOC, scored with the boundary-aware comparator over the same four documents and the same twenty reference values. The last row is the baseline on its own, without the deterministic engine, the rules or the gate, because those are the apparatus it is built to do without; put behind the same merge and gate as the other arms it reaches 14.0 of 20, which measures the apparatus and not the baseline.

Table S15 sets the limitation identified in each strand of the literature against the mechanism by which this study addresses it.

Table S15. Limitations in current work and the mechanisms proposed in this study.

Limitation in current work	Underlying cause	Mechanism proposed in this study
Rule and NER extractors are accurate but tied to a single template	Patterns are anchored to one authority's layout and transfer poorly	A deterministic engine handles the standard table and an LLM engine covers other layouts, under one schema
LLM extraction can report values that are absent from the document	Generative decoding is not constrained by the source text	A source-grounding gate removes any value that cannot be located in the document
Strict structured output is applied by default and rarely measured	Engineers attach JSON schemas without testing their effect	The effect is measured here; schema-less decoding with a normalizer is found to recover the output
Monetary fields are extracted without internal consistency checks	Single-field extractors cannot relate one figure to another	A validation gate tests base plus VAT against the total and flags mismatches
Per-document model cost discourages portfolio-scale use	Frontier models are queried for every field and chunk	A deterministic-first cascade with a light model lowers cost and keeps a free default path

Analysis of Table S15 shows that the proposed mechanisms are individually modest and draw on established ideas; their contribution lies in their combination and in the measurement of their joint effect on real tender documents, which the evaluation reports.

Table S16 lists the fields the workflow populates, grouped by their role in the bid decision, with the reason each group matters to a firm deciding whether to compete.

Table S16. Summary-card fields and their relevance to the bid decision.

Field group	Representative fields	Relevance to the bid decision
Identification	contracting body, object, contract type, procedure	screening for eligibility and relevance
Budget	base budget (with and without VAT), VAT, estimated value	margin feasibility and cash-flow sizing
Schedule	execution period, submission deadline, guarantee period	resource planning and go or no-go timing
Guarantees and insurance	provisional and definitive guarantees, mandatory insurance	bonding cost and locked-up capital
Qualification	economic and technical solvency, business classification	legal eligibility to submit an offer

Analysis of Table S16 shows that the schema is organised around the information a firm needs in order to judge eligibility and to size the cost of a bid, which is the reason the extraction targets these particular fields and validates the relations among the budget figures.

Note S17 states the two comparators in full and defines every quantity the manuscript reports.

Note S17. Scoring criteria and the definition of every reported quantity.

Metric	Definition
Raw field coverage	Number of fields populated, out of 88 (22 fields across four documents), regardless of correctness; a populated-field count
Recall, formerly reported as accuracy	Number of the 20 reference values correctly recovered, over 20; recall over the reference values
Locatability pass rate	Share of surfaced values whose figure or text is located in the normalised source; a property of retrieval and not of correctness
Value failing the locatability test	A surfaced value whose characters cannot be located in the source; an upper bound on fabricated or paraphrased content
Cost per document	Model interface charge for one analysis, computed from measured token counts; zero on the deterministic path. Reading time, deployment and operation are not included
Arithmetic consistency	Whether base plus VAT equals the total and the estimated value is at least the base budget
Precision	Correct values over values emitted, on the 20 reference fields only
F1	Harmonic mean of precision and recall
Abstention	Reference fields on which no value was emitted, over 20
Incorrect emission	Reference fields on which a value was emitted and does not match the reference, over 20
Applicable-field coverage	Populated fields over the fields a document actually contains; not computable here, because no per-document list of applicable fields exists

The permissive comparator is substring containment after a normalisation that decomposes characters, removes diacritics, collapses whitespace and lowercases; it does not touch the full stop or the comma, so a value returned in the Anglo-Saxon convention is not reconciled with a reference written in the Spanish one. It is permissive in one further respect that was measured rather than assumed: ten of the twenty reference values are numeric tokens with no left boundary, so a candidate of 15 percent would be scored correct against a reference of 5 percent, 121 percent against 21 percent, and 124 MESES against 24 MESES. Measured against the one value-level archive of the labelled corpus the defect is latent and not realised: the boundary-aware comparator, which parses percentages, amounts and durations as numbers under token boundaries and compares the remaining fields by whole-word containment, agrees with containment on all seventeen scored fields. The two nonetheless disagree on exactly one labelled slot of the canonical campaign, in every one of the ten runs, and the defect is therefore realised in the opposite direction, as a false rejection. The quantities the manuscript reports are: raw field coverage, the number of fields populated out of 88, regardless of correctness; the locatability pass rate, the share of surfaced values whose figure or text is located in the normalised source, a property of retrieval and not of correctness; a value failing the locatability test, an upper bound on fabricated or paraphrased content; cost per document, the model

interface charge for one analysis, computed from measured token counts and excluding reading time, deployment and operation; arithmetic consistency, whether base plus value added tax equals the total and the estimated value is at least the base budget; precision, correct values over values emitted on the 20 reference fields; F1, the harmonic mean of precision and recall; abstention, the reference fields on which no value was emitted, over 20; and incorrect emission, the reference fields on which a value was emitted and does not match, over 20. Applicable-field coverage, populated fields over the fields a document actually contains, is not computable here, because no per-document list of applicable fields exists: the reference file marks a deferred field as unlabelled rather than as inapplicable, and establishing that list is part of the annotation protocol released with the harness.

Table S18 separates, for each configuration, the fields that are populated, the fields left empty, the reference values that are correct, and the share of surfaced values that fail the locatability test.

Table S18. Breakdown of populated, correct and unlocatable fields, campaign C-R2-4DOC.

Configuration	Populated (/88)	Empty (/88)	Correct of 20	Failing locatability
Deterministic engine	13.0	75.0	13.0	0.0 %
Language model alone (normalizer)	57.0 $\pm$ 2.8	31.0	16.5 $\pm$ 0.5	18.4 %
Full workflow	48.0 $\pm$ 3.2	40.0	17.0	0.0 %

Analysis of Table S18 shows that the language-model engine populates the most fields but that about one in five of its surfaced values fails the locatability test, whereas the full workflow populates fewer fields, surfaces none that fails it and recovers the most reference values. A count of semantically wrong but grounded values over the full eighty-eight fields cannot be given, because only the twenty verified fields have reference values.

Table S19 reports the outcome of removing each component of the workflow in turn and re-scoring the corpus.

Table S19. Effect of removing each component, campaign C-R2-4DOC.

Configuration	Raw coverage of 88	Correct of 20	Locatability pass rate and precision
Full workflow	48.0 $\pm$ 3.2 of 88	17.0 of 20	100.0 %; precision 94.4 %
Remove grounding gate	58.5 $\pm$ 2.3 of 88	19.0 of 20	82.1 %; precision 95.0 %
Remove arithmetic validation	48.0 $\pm$ 3.2 of 88	17.0 of 20	identical to the full workflow in every column
Remove deterministic engine	46.5 $\pm$ 3.7 of 88	14.5 $\pm$ 0.5 of 20	100.0 %; precision 87.9 % $\pm$ 0.4

Analysis of Table S19 supports the reading given in the manuscript: the gate costs exactly 2.00 correct values in every run and buys locatability rather than correctness, the accuracy of the workflow comes from combining the

two engines rather than from either one, and removing the arithmetic validation changes nothing in any column. A fifth configuration, the language-model output consumed without the normalizer, is not a row of this table because it is not a workflow ablation: it measures what a strict consumer obtains from the raw model output, which is nothing, whereas a genuine no-normalizer ablation would retain the deterministic engine and the gate and could not fall below the deterministic floor of 13 of 88. That ablation has still not been run.

Table S20 reports the label-free replication, including coverage across the 572 field slots and the share of surfaced values that could not be located in the source, averaged over the three complete runs of campaign R-POST with 95 percent confidence intervals.

Table S20. Label-free replication on the twenty-six-document corpus, campaign R-POST.

Configuration	Coverage (of 572)	Failing locatability
Deterministic engine only	96 (17 %)	0 %
Language-model engine alone	393.0 $\pm$ 5.0	26.0 $\pm$ 0.0 %
Hybrid, ungated (rules + model)	412.0 $\pm$ 5.0	23.7 $\pm$ 1.4 %
Full workflow (gated)	314.3 $\pm$ 2.9	0 %
Full workflow (relaxed token-based gate, post-hoc)	318.0 $\pm$ 4.3	0 %

Analysis of Table S20 shows that on heterogeneous documents the language-model engine carries the generalisation while the gate preserves the trust guarantee. A fourth complete run on the same configuration, recorded in a campaign that aborted before printing an aggregate block, gives 312 of 572 for the full workflow; including it moves the mean from 314.3 to 313.8. The corresponding pre-repair figures from campaign R-PRE enter the paper only in the comparison of Fig. 4. Value accuracy is not reported for this corpus, since no independent golden annotation exists for it.

Note S21 records the earlier cross-provider evaluation that first indicated that a flat schema the endpoint accepts is preferable to schema-less decoding with a tolerant normalizer.

Note S21. Cross-provider evaluation of the accepted flat schema.

That evaluation was executed on a second provider model, deepseek-chat, and not on the model of the canonical campaign. It returned usable output for all four documents, recovering 51 of 88 field slots with 14 of the 20 reference values correct, against 56 slots and 12 correct values for the schema-less configuration with a tolerant normalizer on the same model. Those two figures are therefore not comparable with Table 2 of the manuscript, whose language-model rows are Gemini 2.5 Flash, and they are reported here as the cross-provider indication that

the controlled arms of campaign C-R2-4DOC then confirmed. The specific HTTP 400 error that the full nested schema provoked is implementation dependent and may differ between providers.

Table S22 sets out the cost calculation for a single document, contrasting the frontier-model baseline with the workflow model. The workflow figures are computed from measured token counts at the provider prices published on 16 September 2026, and will change as those prices change.

Table S22. Per-document cost calculation.

Item	Frontier baseline	Workflow model
Model	frontier reasoning model, not named in any transcript	Gemini 2.5 Flash
Calls per document	one per text chunk	one
Input tokens (approx.)	whole document, repeated	23,029, measured over 40 calls (82,646 characters)
Output tokens (approx.)	record plus reasoning	3,914, measured; no reasoning tokens, the budget being zero
Pricing date	not applicable; the model is not named	16 September 2026, paid standard tier
Token counts	not measured; projected from a published price list	measured on gemini-2.5-flash, campaign C-R2-4DOC
Cost per document	not computed; no published price applies to an unnamed model	USD 0.0167 per document; zero as run, the campaign having used the free tier

Analysis of Table S22 shows that the charge follows from issuing a single call per document to a light model with the input bounded to the document head and the authoritative annex, at 23,029 prompt and 3,914 output tokens per call. Token counts are means over the 40 document calls of the canonical arm and the 20 of the flat-schema arm, read from the usage the interface returns on every call; the flat-schema arm consumes the same 23,029 input tokens and 1,103 output tokens against 3,914, which is USD 0.0097 per document against USD 0.0167. The frontier baseline is a projection: no transcript records which model it used, so no published price can be applied to it and its charge is not computed here. The campaign ran on the free tier and incurred no charge at all. On the deterministic path of the deployed prototype no tokens are consumed, but the harness evaluated here never takes that path.

Note S23 reports the three sensitivities the manuscript summarises: the strictness of the locatability test, the material placed in the model request, and the removal of the apparatus of this paper altogether. They are kept apart from the canonical result because each was designed or identified on the same four documents against which it would then be reported, so each figure is obtained in-sample and cannot be an unbiased estimate. None is used in the abstract or in the conclusions.

Note S23. The three sensitivity analyses in full.

The first sensitivity varies the strictness of the locatability test. The gate originally accepted a textual value only when its leading characters matched the source verbatim, which discarded correct categorical values that the model phrased with additional words, for example a contract type returned as a short phrase where the source carries only the head word. Replacing that prefix test with a token-based one, in which every content word of a textual value must appear in the source while a number must still match verbatim, raises the correct count of the full workflow over the ten runs of the canonical campaign from 17.0 to 18.0 of twenty. The gain is not free: coverage falls from 48.0 plus or minus 3.2 to 45.0 plus or minus 2.1 of 88, because a token-based test admits a value only when every content word occurs in the source. It does not bring the gated workflow level with the ungated combination, which recovers 19.0 of twenty, and it weakens the guarantee it is meant to preserve: the relaxed gate leaves no surfaced value that fails its own locatability test, but that measurement is circular, and measured against the strict test 6.7 percent of the values it surfaces would not pass. Part of the apparent accuracy cost of the gate is therefore an artefact of an over-strict check rather than a failure of extraction, and the test can be relaxed on categorical fields while the verbatim requirement is kept on the fields that carry financial risk. Validating the relaxed gate on labelled documents that played no part in its design remains future work.

The second sensitivity concerns what the model is shown, since a field the selection rule does not send cannot be recovered however capable the model is. The share of annotated evidence that reaches the model was measured on the text layer of the campaign itself and is reported in Table S13. The current rule delivers 19 of the 20 annotated values and sends 330,584 characters over the four documents; the one value it withholds is the definitive guarantee of 5 percent on Pliego C, where the last heading beginning ANEXO I and the following annex heading lie about 330 characters apart, so the rule sends 12,258 characters, 3.2 percent of that document. Supplying the whole cleaned text delivers all 20 annotated values and 1,385,223 characters; field-specific passage selection delivers 19 and 375,099 characters, losing a different one on Pliego A to the truncation cap. Two arms then measure what the model does with the evidence rather than what reaches it. Sending the whole cleaned text raises coverage from 48.0 plus or minus 3.2 to 61.4 plus or minus 0.5 of 88 and correct values from 17.0 to 18.0 of 20, at 88,502 prompt tokens per call against 23,029, nearly four times the input and more than twice the median latency. Cue-driven passage selection does better than either, at 58.6 plus or minus 1.6 of 88 and 19.0 of 20 at 25,495 prompt tokens, within a

tenth of the canonical input size. The head-and-annex rule of the canonical configuration is therefore not the best of the three on either axis, and the paper reports it as the rule the evaluated system uses rather than as a justified choice.

The third sensitivity compares the workflow with a method that shares none of its apparatus. It is a stand-in and not a product: no document extraction system trained for Spanish procurement could be run in this environment without installing a model stack, so the baseline is the same language model as the canonical configuration, prompted zero-shot for generic key-value extraction with no schema, no normalizer, no gate and no rules, and its output projected onto the twenty-two-field schema. What it measures is the value of the apparatus around the model, not the state of the art in document extraction. It recovers 19.0 plus or minus 6.7 of the 88 field slots and 4.6 plus or minus 0.5 of the 20 reference values, against 48.0 plus or minus 3.2 and 17.0 for the workflow. A lenient upper bound that counts a reference value as recovered whenever any emitted pair carries it, under any key, reaches 7.6 of 20, so the projection onto the schema is not what limits it. It is also the only arm of the campaign whose responses were truncated at the output limit, on 5 of its 20 document calls, which leaves its usable-record completion at 65 percent against 100 percent everywhere else. Put behind the same merge and gate as the other arms it reaches 14.0 of 20, which measures the apparatus and not the baseline.

Note S24 carries three examinations that the manuscript states only in summary: the merge rule between the two engines, the per-configuration metrics, and the comparison against the published registry records.

Note S24. The merge rule, the error classes and the registry comparison.

On the twenty reference fields both engines produced a value on 7 occasions, and the deterministic value was kept on all of them. On six the two agree in substance and differ only in formatting, as in 53.169,72 EUR against 53.169,72EUR, or Abierto against Procedimiento Abierto. On the seventh, the tax-inclusive base budget of Pliego A, they disagree: the deterministic engine reads 135.909.110,23 EUR, which is the reference value, and the model returns 164.450.023,38 euros, which is not. Deterministic precedence therefore prevented an error on the one occasion when it mattered and displaced nothing correct, which supports the a priori rule without establishing it beyond these four documents. A conflict flag that surfaces both values, and a field-specific arbitration rule, remain untested. A per-field breakdown released with the harness shows the division of labour directly: the deterministic engine recovers the large monetary fields, for example the budget inclusive of value added tax, which the model alone never recovers on the largest tender, while the model recovers categorical fields such as the contract type and the procedure; the deterministic precedence overrides a model value in seven of the twenty reference fields.

Read as the quantities of Note S17 over the twenty reference fields, precision, recall and F1 are 100.0 percent, 65.0 percent and 78.8 percent for the deterministic engine, 89.2 plus or minus 0.3, 82.5 plus or minus 2.6 and 85.7 plus or minus 1.6 percent for the language model alone, 95.0 percent throughout for the ungated hybrid, and 94.4, 85.0 and 89.5 percent for the full workflow, which abstains on 2.0 of 20 fields and emits one incorrect value. Every one of these quantities was absent from the archived campaign, which recorded how many reference values a configuration recovered but not on how many reference fields it emitted a value at all; the September campaign records emission per field, which is what makes precision computable.

The registry comparison addresses in part the concern that every accuracy figure rests on labels produced by one annotator. The contracting platform publishes a structured record alongside each specification, and where a field appears in both, that record is an independent reference created by the contracting authority and never inspected during development. Table S4 reports the outcome. The budget field is not scored, because the platform publishes three distinct amounts and the harvested manifest retained one of them without recording which. The comparison does not replace independent re-annotation of the twenty verified values, because the platform publishes none of the solvency, classification or guarantee thresholds that motivate this work; its value here was diagnostic, in that it exposed the truncation defect which four labelled documents had not revealed.